Fluid Dynamics II

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1 Introduction

The Part IB Fluid Dynamics course mainly covers inviscid flow and irrotational flow including potential flow. The course covered an introduction to viscous flow for the very simple case where $\mathbf{u} = (u(y,t), 0)$ in 2D parallel flow.

This course is about the effects of viscosity and answers the following very important questione

(i) How does viscosity affect the equations for a general flow?

(ii) What happens if viscosity dominates over the inertia of the fluid?

(iii) What happens if the flow is nearly parallel?

(iv) What happens if viscosity is small compared to inertia?

(v) What about stability?

2 Equations for a Newtonian Viscous Flow

2.1 Dynamic viscosity

Recall from Part IB Fluid Dynamics that we have Couette flow, in which a plate at rest and a plate moving with velocity U are parallel, with a gap h between them filled with a viscous fluid. For a simple fluid, we find that the force required to move the top plate F is proportional to $\frac{UA}{h}$. Hence the shear stress is

$$\tau_s = \frac{F}{A} = \mu \frac{U}{h}$$

where μ is the dynamic viscosity. The shear stress is due to the relative motion between each layer of fluid and its neighbours above and below.

2.2 Rate of strain tensor

Consider the velocity near a fixed point $\mathbf{0}$ *without loss of generality.* Taylor expand $\mathbf{u}$, so we have the multivariable Taylor expansion

$$u_i(\mathbf{x}) = u_i(0) + x_j \left.\frac{\partial u_i}{\partial x_j}\right|_{\mathbf{0}} + \frac{1}{2} x_j x_k \left.\frac{\partial^2 u_i}{\partial x_j \partial x_k}\right|_{\mathbf{0}} + \cdots$$

We drop the quadratic terms and decompose $\frac{\partial u_i}{\partial x_j}$ into symmetric and antisymmetric parts, so

$$\begin{aligned}\frac{\partial u_i}{\partial x_j} &= \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) \\ &= e_{ij} + \frac{1}{2}\varepsilon_{jik}\omega_k\end{aligned}$$

where e_{ij} is the *rate of strain tensor* and $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity. Hence we have that

$$\mathbf{u}(\mathbf{x}) \approx \mathbf{u}(\mathbf{0}) + \mathbf{e} \cdot \mathbf{x} + \frac{1}{2}\boldsymbol{\omega} \times \mathbf{x}.$$

We have that $e_{ij} = e_{ji}$ by definition so $\mathbf{e}$ is real symmetric and hence has real orthogonal eigenvectors, which are the principal axes of strain. For an incompressible flow we have that $e_{ii} = \nabla \cdot \mathbf{u} = 0$ so the eigenvalues sum to 0. The eigenvalues are called the principal rates of strain.

2.3 The stress tensor

Recall in mechanics we consider two types of forces:

(i) Body forces $\mathbf{f}$ that act per unit volume, such as gravity;

(ii) Surface forces $\boldsymbol{\tau}$ that act per unit area, for example, pressure and viscous shear stress

Let $\boldsymbol{\tau}(\mathbf{n}; \mathbf{x}, t)\delta S$ denote the force exerted by the positive side on the negative side of the small area element with $\mathbf{n}$ pointing from the negative side to the positive side.

For example, we have that $\boldsymbol{\tau}(\mathbf{n}) = -p\mathbf{n}$ for an inviscid fluid.

Now let $\mathbf{n}\delta S$ be the sloping face of a small tetrahedron of size ℓ whose three other faces are aligned with the coordinate planes. The force acting on the sloped face is $\boldsymbol{\tau}(\mathbf{n})\delta S$. The forces on the other faces are $\boldsymbol{\tau}(-\mathbf{1})\delta S_1, \boldsymbol{\tau}(-\mathbf{2})\delta S_2, \boldsymbol{\tau}(-\mathbf{3})\delta S_3$. Considering the force balance, the body force and inertia are $O(\ell^3)$ and the surface forces $\boldsymbol{\tau}$ are $O(\ell^2)$, so their ratio is $O(\ell) \to 0$ as $\ell \to 0$. So the surface forces must balance themselves for small ℓ.

Note that pressure and viscous stresses balance inertia and gravity on the characteristic length scale, so small ℓ means that ℓ is small compared to the characteristic length scale L.

Since the forces balance, we have that

$$\boldsymbol{\tau}(\mathbf{n})\delta S + \boldsymbol{\tau}(-\mathbf{1})\delta S_1 + \boldsymbol{\tau}(-\mathbf{2})\delta S_2 + \boldsymbol{\tau}(-\mathbf{3})\delta S_3 = 0.$$

Now

$$\delta S_i = \mathbf{i} \cdot (\mathbf{n}\delta S) = n_i \delta S$$

and $\boldsymbol{\tau}(-\mathbf{1}) = -\boldsymbol{\tau}(\mathbf{1})$ by Newton's Third Law. Substituting these in, we get

$$\boldsymbol{\tau}(\mathbf{n}) = \boldsymbol{\tau}(\mathbf{1})n_1 + \boldsymbol{\tau}(\mathbf{2})n_2 + \boldsymbol{\tau}(\mathbf{3})n_3$$

which shows that $\boldsymbol{\tau}$ is linear in $\mathbf{n}$. So we have that

$$\tau_i(\mathbf{n}) = \sigma_{ij} n_j$$

where $\sigma_{ij} = \tau_i(\mathbf{j})$.

Definition. (Stress tensor) We call $\sigma(\mathbf{x}, t)$, as constructed above, the *stress tensor*.

The key result is that the stress $\boldsymbol{\tau}$ exerted on a small area element (by the outside on the inside) is given by $\boldsymbol{\tau}(\mathbf{n}) = \sigma \cdot \mathbf{n}$.

For the inviscid case $\sigma = -p\,\mathbf{I}$, i.e. $\sigma_{ij} = -p\delta_{ij}$, where $\mathbf{I}$ is the identity tensor.

We will now show that σ is a symmetric tensor.

Consider the angular momentum balance on a small ($\ell \ll L$) arbitrary volume V about a fixed point in V, which we call $\mathbf{0}$.

The moment of the surface forces is

$$\int_{\partial V} \varepsilon_{ijk} x_j \sigma_{km} n_m \mathrm{d}S$$

which is $O(\ell^3)$. Similarly, with ρ the density, the moment of the body forces and inertia is

$$\int_V \mathbf{x} \times \left(\mathbf{f} - \rho \frac{D\mathbf{u}}{Dt} \right) \mathrm{d}V$$

which is also $O(\ell^4)$.

For small ℓ, the moments of the surface forces must balance, hence

$$0 = \int_V \varepsilon_{ijk} \frac{\partial}{\partial x_m}(x_j \sigma_{km}) \mathrm{d}V = \int_V \varepsilon_{ijk} \left(\sigma_{kj} + x_j \frac{\partial \sigma_{km}}{\partial x_m} \right) \mathrm{d}V.$$

Since x_j is $O(\ell)$, we can take $\ell \to 0$ to get rid of the second term, so the integral is approximately

$$\int_V \varepsilon_{ijk} \sigma_{kj} \mathrm{d}V.$$

Since V is arbitrary, $\varepsilon_{ijk}\sigma_{kj} = 0$, hence $\sigma_{ij} = \sigma_{ji}$.

2.4 Constitutive Equation for a Newtonian Fluid

We divide the stress tensor into two terms:

(i) An isotropic pressure term $-p\delta_{ij}$, with $p = -\frac{1}{3}\sigma_{kk}$.

(ii) A deviatoric stress $\sigma_{ij}^{\mathrm{dev}}$ which is the deviation from isotropic and so is traceless: $\sigma_{kk}^{\mathrm{dev}} = 0$.

So we have that

$$\sigma_{ij} = -p\delta_{ij} + \sigma_{ij}^{\mathrm{dev}}.$$

The deviatoric stress is due to frictional interactions with neighbouring fluid particles sliding past with slightly different velocities. For fluid at rest, we must have a static equilibrium where $\nabla p = \mathbf{f}$. Relative motion in the neighbourhood of a point is given by the velocity gradient, so we expect that

$$\sigma^{\mathrm{dev}} = \mathcal{F}[\nabla \mathbf{u}]$$

and $\mathcal{F}(\mathbf{0}) = \mathbf{0}$. A Newtonian fluid is defined by two properties.

(i) σ^{dev} is linear in the local, instantaneous value of $\nabla \mathbf{u}$, hence we have that

$$\sigma_{ij}^{\mathrm{dev}} = A_{ijk\ell} \frac{\partial u_k}{\partial x_\ell}$$

for some tensor $A_{ijk\ell}$.

Local instananeous is good for simple (Newtonian) fluids for which $\ell_{\mathrm{mol}} \sim 10^{-9}$ m and $t_{\mathrm{mol}} \sim 10^{-12}$ s. This can go wrong since we can have action at a distance such as a flow which contains a suspension of fibres. The fluid might also have memory due to deformation elasticity such as a solution contain long-chain polymers.

(ii) The fluid is locally isotropic. This is not true for example in a liquid crystal. So $A_{ijk\ell}$ must be an isotropic tensor, hence

$$A_{ijk\ell} = \mu' \delta_{ij}\delta_{k\ell} + \mu'' \delta_{ik}\delta_{j\ell} + \mu''' \delta_{i\ell}\delta_{jk}.$$

Together this gives us that

$$\sigma_{ij}^{\text{dev}} = \mu' \delta_{ij} \frac{\partial u_k}{\partial x_k} + \mu'' \frac{\partial u_j}{\partial x_i} + \mu''' \frac{\partial u_i}{\partial x_j}$$

and luckily for us the first term is zero since the flow is assumed to be incompressible. Now since σ is symmetric we have that $\mu'' = \mu'''$, so let's drop the primes and call them both μ.

This gives us

$$\begin{aligned}\sigma_{ij} &= -p\delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \\ &= -p\delta_{ij} + 2\mu e_{ij}.\end{aligned}$$

This is called the constitutive relationship for an incompressible Newtonian viscous fluid.

Remark. We'll make some remarks on this relationship.

(i) No dependence on the vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u}$, only depends on the strain-stress rate $e = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$, since localy solid-body rotation $\frac{1}{2}\boldsymbol{\omega} \times \mathbf{x}$ doesn't involve sliding or friction.

(ii) Many common fluids (air, water, oil, syrup) match our assumptions for getting to these equations up to a very close approximation.

(iii) μ here is the same as in IB FLuid Dynamics. Since if we take a flow $\mathbf{u} = (u(y), 0, 0)$ we find that $\sigma_{xy} = \mu \frac{\partial u}{\partial y}$ as we did last year.

2.5 The Momentum Equation

2.5.1 The Cauchy Momentum Equation

The momentum in an arbitrary fixed volume V can change with time due to

(i) Momentum being carried across the boundary ∂V;

(ii) Applying body forces acting on V;

(iii) Applying surface forces on ∂V.

So we have that

$$\frac{d}{dt}\int_V \rho u_i \mathbf{d}V = -\int_{\partial V} \rho u_i (\mathbf{u} \cdot \mathbf{n}) \mathrm{d}S + \int_V f_i \mathrm{d}V + \int_{\partial V} \sigma_{ij} n_j \mathrm{d}S.$$

We'll use that fact that V is fixed and apply the divergence theorem, so

$$\int_V \frac{\partial}{\partial t}(\rho u_i)\mathrm{d}V + \int_V \frac{\partial}{\partial x_j}(\rho u_i u_j)\mathrm{d}V = \int_V \left(f_i + \frac{\partial}{\partial x_j}\sigma_{ij}\right)\mathrm{d}V.$$

Now since V is arbitrary we can drop the integral, so we get

$$u_i\left(\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j)\right) + \rho\left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right) = f_i + \frac{\partial \sigma_{ij}}{\partial x_j}.$$

Notice that the first bracketed term is the left-hand side of the mass conservation equation, so it is zero. This gives us the Cauchy momentum equation,

$$\rho \frac{D\mathbf{u}}{Dt} = \rho\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u}\right) = \mathbf{f} + \nabla \cdot \sigma.$$

2.5.2 The Navier-Stokes Equation

For a Newtonian fluid $\sigma_{ij} = -p\delta_{ij} + 2\mu e_{ij}$. We assume that μ is uniform in the fluid, so we get that

$$\frac{\partial}{\partial x_j}\sigma_{ij} = -\frac{\partial p}{\partial x_j}\delta_{ij} + \mu\left(\frac{\partial^2 u_i}{\partial x_j \partial x_j} + \frac{\partial^2 u_j}{\partial x_j \partial x_i}\right).$$

And we can use the fact that the fluid is incompressible, to get the *Navier-Stokes equation*

$$\rho\frac{D\mathbf{u}}{Dt} = \mathbf{f} - \nabla p + \mu\nabla^2\mathbf{u}.$$

2.5.3 Boundary conditions

For a material boundary with normal $\mathbf{n}$ we have that

(i) Mass conservations gives that $\mathbf{u}\cdot\mathbf{n}$ is continuous;

(ii) $\nabla\mathbf{u}$ is finite because we need σ to be finite, so $\mathbf{u} - (\mathbf{u}\cdot\mathbf{n})\mathbf{n}$ is continuous for a viscous fluid. This is the no slip condition. These together mean that $\mathbf{u}$ is continuous.

(iii) From Newton's Third law we get that $\sigma\cdot\mathbf{n}$ is continuous since the stresses must balance on the boundary. i.e the body stress on the fluid is the negative of the fluid stress on the solid.

2.6 Energy Equation and Viscous Dissipation

Obtain a mechanical energy equation by dotting the Cauchy momentum equation with $\mathbf{u}$ so we get

$$u_i\rho\left(\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j}\right) = u_i f_i + u_i\frac{\partial \sigma_{ij}}{\partial x_j}$$

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\rho u^2\right) - \frac{1}{2}u^2\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x_j}\left(\frac{1}{2}\rho u^2 u_j\right) - \frac{1}{2}u^2\frac{\partial}{\partial x_j}(\rho u_j) = u_i f_i + \frac{\partial}{\partial x_j}(u_i\sigma_{ij}) - \sigma_{ij}\frac{\partial u_i}{\partial x_j}.$$

Now we integrate over a fixed volume V to get the following equation.

Definition. (Energy Equation)

$$\frac{d}{dt}\int_V \frac{1}{2}\rho u^2 \,\mathrm{d}V + \int_{\partial V}\frac{1}{2}\rho u^2(\mathbf{u}\cdot\mathbf{n})\,\mathrm{d}S = \int_V \mathbf{u}\cdot\mathbf{f}\,\mathrm{d}V + \int_{\partial V}\mathbf{u}\cdot\sigma\mathbf{n}\,\mathrm{d}S - \int_V \sigma_{ij}e_{ij}\,\mathrm{d}V.$$

Now we can explain each term in the equation in the order as written.

(i) The rate of change of kinetic energy inside V.

(ii) Kinetic energy being carried across the boundary ∂V.

(iii) The rate of doing work by body forces inside V.

(iv) The rate of doing work by surface forces $\boldsymbol{\tau} = \sigma\cdot\mathbf{n}$ on ∂V.

(v) The rate of doing work against the internal stresses due to deformation.

For incompressible Newtonian fluids $\sigma_{ij} = -p\delta_{ij} + 2\mu e_{ij}$ so we have that

$$\sigma_{ij}e_{ij} = -pe_{ii} + 2\mu e_{ij}e_{ij}$$

and e_{ii} vanishes since the flow is incompressible.

Hence in the case for an incompressible fluid, the fifth term gives the rate of loss of mechanical energy (dissipation) due to viscosity and defomration. The energy lost if converted to heat and is a source term in the temperature equation.

Definition. (Localy viscous dissipation rate) We define Φ as the *viscous dissipation rate* given as

$$\Phi = 2\mu e_{ij}e_{ij}.$$

2.7 Nondimensionalisation

Suppose a flow has a characteristic length scale L and velocity scale U and varies on a timescale T. Assume no body forces so we get that

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u}, \quad \nabla\cdot\mathbf{u} = 0$$

where $\mathbf{u} = \mathbf{u}(\mathbf{x}, t, \rho, \mu, U, L, T, \text{shape}, \dots)$. Define a dimensionless velocity $\mathbf{u}^* = \frac{\mathbf{u}}{U}$, position $\mathbf{x}^* = \frac{\mathbf{x}}{L}$, time $t^* = \frac{t}{T}$, pressure $p^* = \frac{p}{\mu U/L}$ (an alternative inertial scaling sets $p^{**} = \frac{p}{\rho U^2}$).

And substitute these into the Navier-Stokes equation. This gives that

$$\rho\left(\frac{U}{T}\frac{\partial u^*}{\partial t} + \frac{U^2}{L}\mathbf{u}^*\cdot\nabla^*\mathbf{u}^*\right) = -\frac{\mu U}{L^2}\nabla^* p^* + \frac{\mu U}{L^2}\nabla^{*2}\mathbf{u}^*.$$

So collecting constants, we get that

$$\mathrm{Re}\left(\mathrm{St}\frac{\partial u^*}{\partial t^*} + (\mathbf{u}^*\cdot\nabla)\mathbf{u}^*\right) = -\nabla^* p^* + \nabla^{*2}\mathbf{u}^*$$

where we have the following defintions

Definition. (Reynolds number) For a flow with length scale L, velocity scale U, density ρ, and dynamic velocity μ we have

$$\mathrm{Re} = \frac{\rho U L}{\mu} = \frac{UL}{\nu}$$

where we are defining

$$\nu = \frac{\mu}{\rho}$$

to be the *kinematic viscosity.*

We also have $\mathrm{St} = \frac{L}{UT}$ as the *Stroughal number* although this comes up much less.

These are dimensionless quantities which we can use to characterise properties of the flow.

Now we can have $\mathbf{u}^* = \mathbf{u}^*(\mathbf{x}^*, t, \mathrm{Re}, \mathrm{St}, \mathrm{Shape})$ which is a function of fewer parameters. If the flow is steady then we can set $T = \frac{L}{U}$, called the *advection time*, and $\mathrm{St} = 1$, so the Reynolds number is the only important parameter. If the boundary conditions are oscillatory, then set $T = \frac{1}{\omega}$, and then group

$$\mathrm{Re}_T = \mathrm{Re} \cdot \mathrm{St}$$

is called the *unsteady Reynolds number* or the *Stokes number*.

$$\mathrm{Re}_T = \frac{L^2}{\nu T} = \frac{\omega L^2}{\nu}$$

which is a ratio of lengthscales.

If two flows with different U, L, T, have the same geometry and same $\mathrm{Re}, \mathrm{St}, f$ and dimensionless boundary conditions, then they are *dynamically similar* because they give the same dimensionless $\mathbf{u}^*$.

In the limit with $\mathrm{St} \ll 1$ we have slow oscillations and hence we can neglect the $\frac{\partial u^*}{\partial t^*}$ term to leading order which results in quasi-steady behaviour. Otherwise if $\mathrm{St} \gg 1$, we have rapid oscillations and we can neglect $\mathbf{u}^* \cdot \nabla \mathbf{u}^*$ which results in a linearised set of equations.

If instead we have $\mathrm{Re} \ll 1$ the flow is viscously dominated and we neglect the LHS to get Stokes equations. If instead we have that $\mathrm{Re} \gg 1$ the flow is inertially dominated on length scale L which means we can often neglect ${\nabla^*}^2 \mathbf{u}^*$ at most positions which results in the Euler equations, but not near rigid boundaries on lengthscales $\ll 1$, which gives our boundary layer equations.

3 Unidirectional Flows

Now start with the Navier-Stokes equations and have σ and dissipation Φ. Suppose that $\mathbf{u} = (u, 0, 0)$. Incompressibility $\nabla \cdot \mathbf{u} = 0$ which implies that $\frac{\partial u}{\partial x} = 0$ which gives that $(\mathbf{u} \cdot \nabla)\mathbf{u} = 0$ and hence $\frac{D\mathbf{u}}{Dt} = \left(\frac{\partial u}{\partial t}, 0, 0\right)$. Naiver-Stokes equation becomes

$$\begin{aligned} \rho \frac{\partial u}{\partial t} &= f_x - \frac{\partial p}{\partial x} + \mu \nabla_\perp^2 u \\ \mathbf{f}_\perp &= \nabla_\perp p \end{aligned}$$

where $u = u(y, z, t)$ and $\perp$ indicates the components perpendicular to the flow direction. If $\mathbf{f} = \mathbf{0}$ then

$$\frac{\partial p}{\partial y} = \frac{\partial p}{\partial z} = 0$$

and

$$\frac{\partial u}{\partial x} = 0 \implies -\frac{\partial p}{\partial x} = G(t)$$

independent of x.

3.1 Couette Flow

Consider steady flow, driven by motion of the boundaries between two plates, the bottom of which is at rest, and the top plate which moves with velocity V in the positive x direction. Now from the previous equations we have that

$$0 = \mu \frac{\partial^2 u}{\partial y^2}$$

along with boundary conditions $u(0) = 0, u(h) = V$ which gives that $= \frac{V}{h}y$. Now

$$\frac{\partial u_i}{\partial x_j} = \begin{pmatrix} 0 & \gamma & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \implies e_{ij} = \begin{pmatrix} 0 & \frac{\gamma}{2} & 0 \\ \frac{\gamma}{2} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \implies \sigma_{ij} = \begin{pmatrix} -p & \mu\gamma & 0 \\ \mu\gamma & -p & 0 \\ 0 & 0 & -p \end{pmatrix}$$

e.g. the stress exerted by the bottom plate on the fluid is given by $\sigma \cdot \mathbf{n}$ with $\mathbf{n} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$. This is equal to $\begin{pmatrix} -\mu\gamma \\ p \\ 0 \end{pmatrix}$.

We have that $\Phi = 2\mu e_{ij} e_{ij} = 2\mu \left(\left(\frac{\gamma}{2}\right)^2 + \left(\frac{\gamma}{2}\right)^2 \right) = \mu\gamma^2$.

3.2 Plane Poiseuille Flow

Consider two plates both at rest placed at $y = -h$ and $y = h$. Suppose the pressure on the right is p_- and p_+ on the left. The result is steady flow driven by the pressure gradient $\frac{dp}{dx} < 0$. So now we have that

$$\mu \frac{\partial^2 u}{\partial y^2} = \frac{dp}{dx}, \quad u(\pm h) = 0.$$

This gives that $u = -\frac{1}{2\mu}\frac{dp}{dx}(h^2 - y^2)$ and hence the flux per unit length

$$q = \int_{-h}^{h} u \, \mathrm{d}y = -\frac{2h^3}{3\mu}\frac{dp}{dx}.$$

The vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u} = \left(0, 0, -\frac{1}{\mu}\frac{dp}{dx}y\right)$ and the strain rate is $\begin{pmatrix} 0 & \frac{1}{2}\frac{\partial u}{\partial y} & 0 \\ \frac{1}{2}\frac{\partial u}{\partial y} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ with dissipation $\Phi(y) = \mu \left(\frac{\partial u}{\partial y}\right)^2$.

Remark. Check the stress exerted *by* the boundary *on* the fluid at $y = \pm h$ is $\left(h\frac{dp}{dx}, \mp p, 0\right)$ and check the total dissipation $\int_V \mathbf{\Phi}(y) \, \mathrm{d}V$ is equal to the rate of working by the stresses on ∂V which is $\int_{\partial V} \mathbf{u} \cdot \sigma \cdot \mathbf{n} \, \mathrm{d}S$.

3.3 Circular Poiseuille Flow

Consider Poiseuille flow in a cylinder with radius a, bottom and top at $z = 0$ and $z = L$ with pressure $p_0 + \Delta P$ at the bottom and p_0 at the top. This pressure gradient induces a flow upwards. We can ignore entrance and exist effects assuming that the cylinder is long and thin so $L \gg a$.

In cylindrical polar coordinates, (r, θ, z) Navier-Stokes equation becomes

$$\frac{\partial p}{\partial r} = \frac{1}{r}\frac{\partial p}{\partial \theta} = 0$$
$$0 = -\frac{\partial p}{\partial z} + \mu \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right) \quad \text{for } \mathbf{u} = (0, 0, w).$$

So since $\nabla \mathbf{u} = 0 \implies \frac{\partial w}{\partial z} = 0 \implies -\frac{\partial p}{\partial z} = G$ and we get

$$\implies \frac{1}{r}\frac{\partial}{\partial r} r \frac{\partial w}{\partial r} = -\frac{G}{\mu} \implies \frac{\partial w}{\partial r} = -\frac{Gr}{2\mu} + \frac{A}{r}$$

and we get that $A = 0$ since we have regularity at $r = 0$ which is part of the flow. Now we get that

$$w = \frac{G}{4\mu}\left(a^2 - r^2\right)$$

which gives our upwards flow, using the no-slip condition on the walls. Similarly we can calculate the flux,

$$\begin{aligned}\text{Volumn Flux} = Q = \int_0^{2\pi}\int_0^a w \, \mathrm{d}r\mathrm{d}\theta &= 2\pi \frac{G}{4\mu}\left[\frac{a^2 r^2}{2} - \frac{r^4}{4}\right]_0^a \\ &= \frac{\pi G a^4}{8\mu}.\end{aligned}$$

We have that

$$\mathbf{u} = w\mathbf{e}_z \implies e = \begin{pmatrix} 0 & 0 & \frac{1}{2}\frac{\partial w}{\partial r} \\ 0 & 0 & 0 \\ \frac{1}{2}\frac{\partial w}{\partial r} & 0 & 0 \end{pmatrix}, \quad \sigma_{rz} = 2\mu\sigma_{rz} = \mu\frac{\partial w}{\partial r} = -\frac{Gr}{2}.$$

We can show that the total shear stress *by* the fluid *on* the tube is equal to the pressure force applied to the fluid which is $\Delta p \pi a^2 \mathbf{e}_z$. The total dissipation is equal to the rate of doing work (Example Sheet 1, Question 5).

Example. We could extend this to flow in an annulus, with two fluids arranged concentrically, or as an ellipsoid.

3.4 Pressure driven flow in a rectangular channel

Consider a rectangle of length L of fluid with flow in the z direction with $\mathbf{u} = w(x, y)\mathbf{e}_z$. Let the input channel be width $2a$ in the x direction and height $2b$ in the y direction. We want

$$\mu\nabla^2 w = \frac{\partial p}{\partial z} = -G$$

with $w = 0$ on $x = \pm a$, $y = \pm b$. Suppose that $a \gg b$. Then we'd have our normal parallel plate flow in the limit

$$w = \frac{G}{2\mu}(b^2 - y^2)$$

which we'll use as our particular solution since it solves the Poisson equation and the boundary condition $w = 0$ on $y = \pm b$, we just need to add a complementy solution for the boundary conditions. So we need a $w^{\mathrm{c}}(x, y)$ such that $\nabla^2 w^{\mathrm{c}} = 0$ which satisfies the boundary conditions. We need $w^c = 0$ at $y = \pm b$ and $w^c = -\frac{G}{2\mu}(b^2 - y^2)$ on $x = \pm a$.

Since Laplace's equation is homogeneous, we can use seperation of variables and Fourier series of the x boundary condition to get that

$$w^{\mathrm{c}} = \frac{Gb^2}{2\mu}\sum_{n=0}^{\infty}\frac{4(-1)^n}{\lambda_n^3}\cos\frac{\lambda_n y}{b}\frac{\cosh\frac{\lambda_n x}{b}}{\cosh\frac{\lambda_n a}{b}}$$

where we're scaling so our eigenvalues of the operator are $\lambda_n = \left(n + \frac{1}{2}\right)\pi$.

So the flux is

$$Q = \int_{-b}^{b}\int_{-a}^{a} w \, \mathrm{d}x\mathrm{d}y = \frac{4\Delta p}{3\mu L} a b^3 \left(1 - \frac{6b}{a}\sum_{n=0}^{\infty} \frac{\tanh\left(\lambda_n \frac{a}{b}\right)}{\lambda_n^5}\right)$$
$$= 2a \cdot q_{2\mathrm{D}} \left(1 - \text{correction due to the sides}\right).$$

3.5 Implusively started flat plate

Consider a fluid in $y > 0$, initially at rest and set into motion by a rigid plate at $y = 0$, which starts to move with velocity $(U, 0)$ at $t = 0$. This results in unidirectional flow $\mathbf{u} = (u(y,t), 0, 0)$ which obeys the equation

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2}, \quad u(0,t) = U$$

for $t > 0$, and $u \to 0$ as $y \to \infty$. The only relevant lengthscale from the equation is $\sqrt{\nu t}$. Our similarity solution is $u = U\mathrm{erfc}(\eta)$ where $\eta = \frac{y}{2\sqrt{\eta t}}$, and $\mathrm{erfc}(\eta) = \frac{2}{\sqrt{\pi}}\int_\eta e^{\infty} e^{-s^2}\, \mathrm{d}s$. The stress by the fluid on the plate is $\mu\frac{\partial u}{\partial y} = -\frac{\mu U}{\sqrt{\pi\nu t}}$. The rate of working by the plate on the fluid is $U\left(-\mu\frac{\partial u}{\partial y}\right)$ which must equal the rate of change of kinetic energy of the flow plus the rate of dissipation.

It's easiest to check from integrating the equation $\rho u\left(\frac{\partial u}{\partial t}\right) = \rho u\left(\nu\frac{\partial^2 u}{\partial y^2}\right)$ which implies that

$$\frac{d}{dt}\int_0^\infty \frac{1}{2}\rho u^2 \, \mathrm{d}y = \mu\int_0^\infty u\frac{\partial^2 u}{\partial y^2}\, \mathrm{d}y = \left[\mu u\frac{\partial u}{\partial y}\right]_0^\infty - \mu\int_0^\infty \left(\frac{\partial u}{\partial y}\right)^2 \, \mathrm{d}y.$$

3.6 Oscillating flat plate

Consider a fluid in $y > 0$ with a rigid plate at $y = 0$ which has velocity $U\cos(\omega t)$ with vertical length scale δ. Then we have

$$\frac{\partial u}{\partial t} = \nu\frac{\partial^2 u}{\partial y^2}$$

again. We have by the no-slip condition that $u = \mathrm{Re}(Ue^{i\omega t})$ at $y = 0$ and the boundary condition that $u \to 0$ as $y \to \infty$. Steady oscillations (after decay of transients gives that

$$u = \mathrm{Re}\left(Ue^{i\omega t}e^{-\sqrt{\frac{i\omega}{\nu}}y}\right)$$
$$= Ue^{-ky}\cos(\omega t - ky)$$

where $k = \left(\frac{\omega}{2\nu}\right)^{\frac{1}{2}}$, so we get that

$$\delta \sim \frac{1}{k} \sim \left(\frac{\nu}{\omega}\right)^{\frac{1}{2}} \sim (\nu T)^{\frac{1}{2}}.$$

Again, the rate of working is given by the change in kinetic energy plus the rate of dissipation, but time dependency makes this complicated.

For the time-averaged energy calculations (quadratic quantities) note that $\langle\cos^2\rangle = \langle\sin^2\rangle = \frac{1}{2}$, $\langle\cos\sin\rangle = 0$, and

$$\langle\mathrm{Re}(Ae^{i\omega t})\mathrm{Re}(Be^{i\omega t})\rangle = \frac{1}{2}\mathrm{Re}(AB^*).$$

3.7 Taylor-Couette Flow

This is flow between long coaxial rotating cylinders. The flow is axisymmetric so $\mathbf{u} = v(r)\mathbf{e}_\theta$. The inner cylinder $r == a$ rotates with angular velocity Ω_2 and the outer cylinder $r = b$ rotates with angular velocity Ω_1. We want Navier-Stokes equations in cylindrical polar coordinates. In fluids we need $\nabla^2\mathbf{u}$. We can use the identity that

$$\nabla^2\mathbf{u} = \nabla(\nabla\cdot\mathbf{u}) - \nabla\times(\nabla\times\mathbf{u})$$

and the first term on the RHS is zero in an incompressible fluid which makes things much simplier. Formula are not required to be memorised for the exam. The case of Taylor-Couette flow however is sufficiently simple so we can do it by hand. We obtain that

$$(\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{v^2}{r}\mathbf{e}_r$$

which is the formula for circular motion. There's no radial component of the viscous term so we get that

$$-\rho\frac{v^2}{r} = -\frac{\partial p}{\partial r}$$

so the radial pressure gradient balances the centrifugal force. In the azimuthal direction (since $\frac{\partial p}{\partial\theta} = 0$) we have that

$$0 = \mu\left(\frac{\partial^2 v}{\partial r^2} + \frac{1}{r}\frac{\partial v}{\partial r} - \frac{v}{r^2}\right)$$

with $v(a_1) = \Omega_1 a_1$ and $v(a_2) = \Omega_2 a_2$. This is a Cauchy-Euler equation, which is solved with the boundary conditions (Example Sheet 1, Question 6).

4 Stokes Flows

Stokes flows are flows with $\mathrm{Re} \ll 1$, so we neglect inertial forces. Then Navier-Stokes equations become

$$0 = \mathbf{f} - \nabla p + \mu\nabla^2\mathbf{u}, \quad \nabla\cdot\mathbf{u} = 0.$$

Recall that $\mathrm{Re} = \frac{UL}{\nu}$ so this is good for U and L small (for example micro-organisms and aerosols) or if the viscosity is very large like lava flows or ice-sheets.

We will often assume that $\mathbf{f} = \mathbf{0}$ as we can remove $\rho\mathbf{g}$ from the equation by subtracting the hydrostatic pressure from the problem, $p_0 - \rho g z$ from p. In this case we have that

$$\mathbf{0} = -\nabla p + \mu\nabla^2\mathbf{u} = \nabla\cdot\sigma.$$

4.1 Simple Properties

(i) The flow is instanteous, so there is no $\frac{\partial}{\partial t}$ term, so this is a pure spacial problem. The flow only depends on the boundary conditions (and $\mathbf{f}$) at each instant. It may depend on time parametrically, in which we say the flow is quasi-steady if for example the boundary conditions are changing with time.

(ii) The equation is linaer since there is no $(\mathbf{u}\cdot\nabla)\mathbf{u}$ term. So $\mathbf{u}, p, \sigma$ are linear in the boundary conditions (and in $\mathbf{f}$).

Example. Suppose we have some body in a fluid moving with velocity $\mathbf{U}$ with a drag force equal to $-A \cdot \mathbf{U}$ where A is a second rank tensor. We can use symmetric/anti-symmetric decomposition to break up a problem into two problems where one velocity is parallel with the axis of the body, and the other is perpendicular.

(iii) Changing the sign of all of the boundary conditions (and $\mathbf{f}$) results in just changing the sign of $\mathbf{u}$. Reversing all the boundary conditions and the history of their application results in the fluid particles retracing their paths exactly.

(iv) If we have symmetry in space then due to linearity and reversibility we can use the symmetry of the geometry to show something is zero.

Example. Consider a sphere falling under $\mathbf{g}$ by a vertical wall. We'll show that the sphere must fall vertically. Take coordinates with z vertical and x perpendicular to the wall. Suppose not and the sphere falls non-vertically with a component in the $x-y$ plane going away from the wall (wlog). Then reverse gravity so we reverse the velocity $\mathbf{U}$. The problem is symmetrical so we can reflect the problem along the sphere's equator, perpendicular to the wall. So we can $\mathbf{g}$ acting down again and $\mathbf{U}$ pointing towards the wall. But by symmetry it can't have a component both towards the wall and away from the wall. So $U_x = 0$ and similarly by a symmetric argument $U_y = 0$.

We'll show a similar result with rotation (psuedovectors) instead of velocity.

Example. (Falling uniform rod does not rotate). Suppose a rod falls with angular velocity Ω due to the falling motion. Reverse g so Ω now goes the other way and g points up. Now rotate by π so we have the same problem with Ω going the opposite direction, so by symmetry we have that $\Omega = 0$, i.e there is no rotation induced by the fall.

(v) We have harmonic properties (for $\mathbf{f} = \mathbf{0}$), so

$$\mu \nabla^2 \mathbf{u} = \nabla p, \quad \nabla \cdot \mathbf{u} = 0$$

so taking divergence of the left equation we get that p is harmoinc. Taking the curl of the left equation we get $\nabla^2 \boldsymbol{\omega} = 0$ so vorticity is harmonic. From taking the Laplacian of the left equation we get that $\nabla^2 \nabla^2 \mathbf{u} = \nabla^4 \mathbf{u} = 0$ which we say that $\mathbf{u}$ is biharmonic.

4.2 More Properties

We'll first assume that $\mathbf{f} = \mathbf{0}$ so we have that $\nabla \cdot \sigma = 0$.

We have uniqueness of flows. Suppose $\mathbf{u}_1$ and $\mathbf{u}_2$ are Stokes flows in a domain V and satisfy the same boundary conditions on ∂V. Let $\mathbf{u} = \mathbf{u}_1 - \mathbf{u}_2$ and $e = e_1 - e_2$. Then we get that

$$\begin{aligned} D = 2\mu \int_V e_{ij} e_{ij} \, \mathrm{d}V &= \int (\sigma_{ij} + p\delta_{ij}) e_{ij} \, \mathrm{d}V \\ &= \int_V \sigma_{ij} \frac{\partial u_i}{\partial x_j} \, \mathrm{d}V \\ &= \int_V \frac{\partial}{\partial x_j}(u_i \sigma_{ij}) - u_i \frac{\partial}{\partial x_j} \sigma_{ij} \, \mathrm{d}V \\ &= \int_{\partial V} \mathbf{u} \cdot \sigma \cdot \mathbf{n} \, \mathrm{d}S \\ &= 0 \end{aligned}$$

since $\mathbf{u}_1 = \mathbf{u}_2$ or the stress equals on the boundary ($\sigma_1 \cdot \mathbf{n} = \sigma_2 \cdot \mathbf{n}$ on ∂V. So we get $\mathbf{e} = 0$ in V. So $\mathbf{u}$ is at most a solid body motion. If $\mathbf{u}_1 = \mathbf{u}_2$ on ∂V them $\mathbf{u}_1 = \mathbf{u}_2$ in V. Instead if $\sigma_1 \cdot \mathbf{n} = \sigma_1 \cdot \mathbf{n}$ then $\mathbf{u}_1 = \mathbf{u}_2 + \mathbf{U} + \mathbf{\Omega} \times \mathbf{x}$.

Theorem. (Minimum dissipation theorem) Let $\mathbf{u^S}$ be a Stokes flow in V and $\mathbf{u}$ be any incompressible flow that has the same velocity on ∂V. Then

$$2\mu \int_V e_{ij}^{\mathrm{S}} e_{ij}^{\mathrm{S}} \, \mathrm{d}V \leq 2\mu \int_V e_{ij} e_{ij} \, \mathrm{d}V$$

Proof. We have that

$$\begin{aligned} 0 &\leq 2\mu \int_V (e_{ij} - e_{ij}^{\mathrm{S}})(e_{ij} - e_{ij}^{\mathrm{S}}) \, \mathrm{d}V \\ &= 2\mu \int_V (e_{ij} e_{ij} - e_{ij}^{\mathrm{S}} e_{ij}^{\mathrm{S}}) \, \mathrm{d}V + 4\mu \int_V e_{ij}^{\mathrm{S}} (e_{ij}^{\mathrm{S}} - e_{ij}) \, \mathrm{d}V. \end{aligned}$$

The last term (using incompressibility in both flows) is equal to

$$\begin{aligned} 2\int_V \sigma_{ij}^{\mathrm{S}} (e_{ij}^{\mathrm{S}} - e_{ij}) &= 2\int_V \frac{\partial}{\partial x_j}((u_i^{\mathrm{S}} - u_i)\sigma_{ij}^{\mathrm{S}}) - (u_i^{\mathrm{S}} - u_i)\frac{\partial}{\partial x_j}\sigma_{ij}^{\mathrm{S}} \, \mathrm{d}V \\ &= 2\int_{\partial V} (u_i^{\mathrm{S}} - u_i)\sigma_{ij} n_j \, \mathrm{d}S = 0 \end{aligned}$$

so we get the required result. □

Example. We can bound drag on a translating rigid body. Consider an arbirary rigid body, translating with velocity $\mathbf{U}$ with Stokes flow $\mathbf{u}^{\mathrm{S}}$ in the fluid volume V around it. Take $\mathbf{n}$ into the boundy so

$$\int_{\partial V} \sigma \cdot \mathbf{n} \, \mathrm{d}S = \text{force } F_{\mathrm{S}} \text{ exerted by body on fluid}$$

we have that the dissipation

$$\begin{aligned} D &= 2\mu \int_V e_{ij}^{\mathrm{S}} e_{ij}^{\mathrm{S}} \, \mathrm{d}V \\ &= \int_{\partial V} \mathbf{u}^{\mathrm{S}} \cdot \sigma \cdot \mathbf{n} \, \mathrm{d}S = \mathbf{U} \cdot \int_{\partial V} \sigma^{\mathrm{S}} \cdot \mathbf{n} \, \mathrm{d}S = \mathbf{U} \cdot \mathbf{F}^{\mathrm{S}}. \end{aligned}$$

Now suppose that S is a sphere of radius a which circumscribes the body. Define a flow $\mathbf{u}$ in the original fluid volume V by taking the Stokes flow outside the rigid sphere S moving the velocity $\mathbf{U}$ and taking uniform flow $\mathbf{U}$ in the gap between S and the body. So we get that

$$2\mu \int_V e_{ij} e_{ij} \, \mathrm{d}V = 2\mu \int_{\text{outside of}} S e_{ij} e_{ij} \, \mathrm{d}V$$

since $e_{ij} = 0$ in the gap. Now this is equal to

$$\mathbf{U} \cdot (\text{force exerted by sphere})$$

which is $6\pi\mu a\mathbf{U}$ (result derivation later). Hence the minimium dissipation gives that

$$\mathbf{U} \cdot \mathbf{F}^{\mathrm{S}} \leq 6\pi\mu a U^2.$$

We have a similar argument for an inscribed sphere of radius b so we get the same formula but with b instead of a. Then for a cube of size $2L$ set $b = L$ and $a = \sqrt{3}L$ so from the symmetries of the cube with $\mathbf{F}^{\mathrm{S}} = A\mathbf{U}$ has the same A for all orientations (Example Sheet 2, Question 2) and we deduce that

$$1 \leq \frac{A}{6\pi\mu L} \leq \sqrt{3}$$

and we have a regular argument for couple and rotation.

Theorem. (Reciprocal Theorem) Let $\mathbf{u}^1$ and $\mathbf{u}^2$ be Stokes flows in some domain V with different boundary conditions on ∂V. Then

$$\int_{\partial V} \mathbf{u}^1 \cdot \sigma^2 \cdot \mathbf{n} \, \mathrm{d}S = \int_{\partial V} \mathbf{u}^2 \cdot \sigma^1 \cdot \mathbf{n} \, \mathrm{d}S.$$

Proof. We'll go over a quick outline of the proof now. Consider

$$2\mu \int_V e^1_{ij} e^2_{ij} \, \mathrm{d}V = \int_V (\sigma^1_{ij} + p^1 \delta_{ij}) e^2_{ij} \, \mathrm{d}V$$

and we can get rid of the pressure term since $\nabla \cdot \mathbf{u}^2 = 0$, and using the fact that $\sigma^1_{ij} = \sigma^2_{ji}$, so we have that (after applying integration by parts and using the fact that the divergence of σ^1 is zero and using the divergence theorem)

$$\cdots = \int_{\partial V} \mathbf{u}^2 \cdot \sigma^1 \cdot \mathbf{n} \, \mathrm{d}S = RHS$$

and since the first line was symmetric, the theorem follows. □

Example. Suppose that $\mathbf{u}^{1,2}$ are Stokes flows due to rigid body motion of a certain body (some V) i.e that $\mathbf{u}^{1,2} = \mathbf{U}^{1,2} + \boldsymbol{\Omega}^{1,2} \times \mathbf{x}$ on ∂V where $\mathbf{U}^i, \boldsymbol{\Omega}^i$ are constants. The reciprocal theorem gives that

$$\mathbf{U}^1 \cdot \mathbf{F}^2 + \boldsymbol{\Omega}^1 \cdot \mathbf{G}^2 = \mathbf{U}^2 \cdot \mathbf{F}^1 + \boldsymbol{\Omega}^2 \cdot \mathbf{G}^1 \tag{*}$$

where

$$\mathbf{F} = \int_{\partial V} \sigma \cdot \mathbf{n} \, \mathrm{d}S \quad \text{and} \quad \mathbf{G} = \int_{\partial V} \mathbf{x} \times \sigma \cdot \mathbf{n} \, \mathrm{d}S$$

are the corresponding forces and couples exerted by/on the body $\mathbf{n}$ out/in of the fluid.

By the linearity of the Stokes equations we have that

$$\mathbf{F} = A \cdot \mathbf{U} + B \cdot \boldsymbol{\Omega}$$
$$\mathbf{G} = C \cdot \mathbf{U} + D \cdot \boldsymbol{\Omega}$$

where the tensors A, B, C, D depend on the shape of the body. We can show that using $(*)$ that $A = A^\top, D = D^\top$ and $B = C^\top$.

For a sphere and a cube, symmetries imply that

$$A = AI, D = DI, B = C^\top = 0$$

where the LHS is a tensor and the RHS is a constant multiplied by the identity tensor.

For a helix $B = C^\top \neq 0$.

4.3 Flow Past a Sphere

4.3.1 Stationary Sphere in Uniform Flow

Consider a sphere with radius a with a uniform flow $\mathbf{U}$ past a fixed rigid sphere, radius a. There are several methods, all which have heavy algebra somewhere or depend on heavy usage of spherical polar coordinates.

The solution is

$$\mathbf{u} = \mathbf{U}\left(1 - \frac{3a}{4r} - \frac{a^3}{4r^3}\right) + \mathbf{x}(\mathbf{U}\cdot\mathbf{x})\left(-\frac{3a}{4r^3} + \frac{3a^3}{4r^5}\right)$$

which we can derive in many painful ways.

4.3.2 Method 1 (semi-modern, linearity and vector method)

From the linearity of the Stokes equations, $\mathbf{u}(\mathbf{x})$ must be linear in $\mathbf{U}$. Taking the origin to be at the centre of the sphere, the spherically symmetric geometry means the only other vector ingredient for $\mathbf{u}$ is $\mathbf{x}$. Hence the velocity and pressure fields must take the forms

$$\begin{aligned}\mathbf{u}(\mathbf{x}) &= \mathbf{U}f(r) + \mathbf{x}(\mathbf{U}\cdot\mathbf{x})\,g(r),\\ p(\mathbf{x}) &= \mu(\mathbf{U}\cdot\mathbf{x})\,h(r),\end{aligned}$$

where $r = |\mathbf{x}|$, and f, g and h are scalar functions of r to be determined.

Differentiate

$$\frac{\partial u_i}{\partial x_j} = U_i x_j f'/r + \delta_{ij}U_n x_n g + x_i U_j g + x_i x_j U_n x_n\, g'/r. \qquad (*)$$

Contract i with j and apply incompressibility:

$$\nabla\cdot\mathbf{u} = U_n x_n\big(f'/r + 4g + rg'\big) = 0.$$

Differentiate again:

$$\begin{aligned}\mu\nabla^2 u_i &= \mu U_i\big(f'' + 2f'/r + 2g\big) + \mu x_i U_n x_n\big(g'' + 6g'/r\big),\\ \nabla_i p &= \mu U_i h + \mu x_i U_n x_n\, h'/r.\end{aligned}$$

In $\mu\nabla^2\mathbf{u} = \nabla p$ equate the bits with U_i and those with $x_i U_n x_n$. So the Stokes equations give

$$f'/r + 4g + rg' = 0, \qquad f'' + 2f'/r + 2g = h \qquad \text{and} \qquad g'' + 6g'/r = h'/r.$$

Eliminating h and then f yields

$$r^2 g''' + 11rg'' + 24g' = 0.$$

This ODE is equidimensional in r so look for solutions of the form $g = r^\alpha$. Substituting, we find $\alpha = 0, -3$ and -5, with associated $f = -(\alpha+4)\,r^{\alpha+2}/(\alpha+2)$ and $h = -(\alpha+5)(\alpha+2)\,r^\alpha$.

Hence the general solution of the assumed form linear in $\mathbf{U}$ is

$$\begin{aligned}\mathbf{u}(\mathbf{x}) &= \mathbf{U}\Big(-2Ar^2 + B + Cr^{-1} - \tfrac{1}{3}Dr^{-3}\Big) + \mathbf{x}(\mathbf{U}\cdot\mathbf{x})\Big(A + Cr^{-3} + Dr^{-5}\Big),\\ p(\mathbf{x}) &= \mu(\mathbf{U}\cdot\mathbf{x})\Big(-10A + 2Cr^{-3}\Big).\end{aligned}$$

Find $\mathbf{e}$ from $(*)$ and thus $\boldsymbol{\sigma}$: the stress across a spherical surface with normal $\mathbf{n} = \mathbf{x}/r$ is

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \mu \mathbf{U}\Big(-3Ar + 2Dr^{-4}\Big) + \mu \mathbf{x}(\mathbf{U} \cdot \mathbf{x})\Big(9Ar^{-1} - 6Cr^{-4} - 6Dr^{-6}\Big).$$

Applying the boundary conditions $\mathbf{u} = \mathbf{0}$ on the rigid sphere and $\mathbf{u} \to \mathbf{U}$ in the far field, we find the coefficients

$$A = 0, \qquad B = 1, \qquad C = -\tfrac{3}{4}a \qquad \text{and} \qquad D = \tfrac{3}{4}a^3,$$

so

$$\mathbf{u} = \mathbf{U}\left(1 - \frac{3a}{4r} - \frac{a^3}{4r^3}\right) + \mathbf{x}(\mathbf{U} \cdot \mathbf{x})\left(-\frac{3a}{4r^3} + \frac{3a^3}{4r^5}\right),$$
$$p = -\frac{3a\mu\, \mathbf{U} \cdot \mathbf{x}}{2r^3} \qquad \text{and} \qquad \boldsymbol{\sigma} \cdot \mathbf{n}\Big|_{r=a} = \frac{3\mu}{2a}\mathbf{U}.$$

Hence the drag on the sphere is

$$\int_{r=a} \boldsymbol{\sigma} \cdot \mathbf{n}\, \mathrm{d}S = 4\pi a^2\, \frac{3\mu}{2a}\mathbf{U} = 6\pi\mu a\, \mathbf{U}.$$

4.3.3 Method 2 (classical, separable streamfunction in spherical r, θ, ϕ)

Use a Stokes streamfunction $\Psi(r, \theta)$ for the axisymmetric flow with

$$u_r = \frac{1}{r^2 \sin\theta}\frac{\partial \Psi}{\partial \theta} \qquad \text{and} \qquad u_\theta = -\frac{1}{r \sin\theta}\frac{\partial \Psi}{\partial r}$$

(from $\mathbf{u} = \nabla \times \mathbf{A}$ and $\mathbf{A} = \Psi\, \mathbf{e}_\phi / r \sin\theta$ so that $\mathbf{u} \cdot \nabla\Psi = 0$). In Stokes flow $\nabla^2 \boldsymbol{\omega} = \mathbf{0} \Rightarrow \nabla^4 \mathbf{A} = (\nabla\times)^4 \mathbf{A} = \mathbf{0}$ and (with patience or Wikipedia)

$$D^2 D^2 \Psi = 0 \qquad \text{where} \qquad D^2 = \frac{\partial^2}{\partial r^2} + \frac{\sin\theta}{r^2}\frac{\partial}{\partial\theta}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\right).$$

The uniform flow at infinity (the nonzero condition) has $\Psi = \frac{1}{2}Ur^2 \sin^2\theta$; try $\Psi = F(r)\sin^2\theta$ and find $F = Ar^4 + Br^2 + Cr + D/r$. Then $A = 0$, $B = \frac{1}{2}$, and apply the BCs on sphere.

4.3.4 Method 3 (semi-classical, hybrid)

The vorticity is a harmonic function (as is the pressure), decaying at infinity, linear in $\mathbf{U}$ and otherwise constructed only from $\mathbf{x}$ by spherical symmetry. Can be shown the decaying solutions of $\nabla^2(\cdot) = 0$ are $\dfrac{1}{r}$, $\nabla\dfrac{1}{r} = -\mathbf{x}/r^3$, $\nabla\nabla\dfrac{1}{r}$, etc. Hence the vorticity, a pseudo vector, (and the pressure, a scalar) must have the form

$$\boldsymbol{\omega} = A\, \mathbf{U} \times \mathbf{x}/r^3 \qquad \big(\text{and } p = \mu B\, \mathbf{U} \cdot \mathbf{x}/r^3\big).$$

Return to spherical polars and solve $D^2\Psi = AU\sin^2\theta/r$ with $\Psi = F(r)\sin^2\theta$. From $\nabla p = -\mu\nabla \times \boldsymbol{\omega}$ get $B = \mu A$.

4.3.5 Method 4 (modern, Papkovich–Neuber)

If $\boldsymbol{\phi}(\mathbf{x})$ is a vector harmonic function ($\nabla^2\boldsymbol{\phi} = \mathbf{0}$) then

$$\mathbf{u} = \nabla(\mathbf{x}\cdot\boldsymbol{\phi}) - 2\boldsymbol{\phi}, \qquad p = 2\mu\,\nabla\cdot\boldsymbol{\phi}.$$

is a solution of the Stokes equations (Sheet 2, Q5). It's not quite the general solution (see Part III), but does include this problem.

Linearity and spherical symmetry give (for the perturbation $\mathbf{u} - \mathbf{U}$, which decays at infinity)

$$\boldsymbol{\phi} = A\,\mathbf{U}\,\frac{1}{r} + B\,\mathbf{U}\cdot\nabla\nabla\frac{1}{r},$$

with coefficients to be determined by applying the boundary conditions.

Remark. Velocity is a true vector. Vorticity/angular velocity are pseudo vectors. (Difference is behaviour under reflections.) Sometimes linearity is not sufficient, you need to pay attention to the true vs. pseudo nature of the vectors involved. E.g. rotation of a sphere: $\mathbf{u}$ (true vector) linear in $\boldsymbol{\Omega}$ (pseudo vector) gives (by Method 1)

$$\mathbf{u} = \boldsymbol{\Omega}\times\mathbf{x}\,f(r).$$

Remark. Often deal with a sphere moving with velocity $\mathbf{U}$ in fluid otherwise at rest. Remove uniform flow from the above, and put $\mathbf{U}\to-\mathbf{U}$. Better, start from scratch with correct BCs. Can also use general algebra (A, B, C, D) with different BCs for a spherical bubble (Example Sheet 2, Question 6), or a spherical drop.

4.3.6 Sphere translating through fluid otherwise at rest

We find that (from previous sections methods) that

$$\mathbf{u} = \mathbf{U}\left(\frac{3a}{4r} + \frac{a^3}{4r^3}\right) + (\mathbf{U}\cdot\mathbf{x})\mathbf{x}\left(\frac{3a}{4r^3} - \frac{3a^3}{4r^5}\right)$$

and

$$p = \frac{3\mu a}{2}\frac{\mathbf{U}\cdot\mathbf{x}}{r^3}, \quad \sigma\cdot\mathbf{n}\Big|_{r=a} = -\frac{3\mu}{2a}\mathbf{U}$$

and the fluid drag on the sphere is $-6\pi\mu a\mathbf{U}$ which is minus the force $\mathbf{F}$ exerted by the sphere on the fluid.

For example consider a falling sphere with no inertia. Then the weight-buoyancy is equal to $\frac{4}{3}\pi a^3(\rho_s-\rho)\mathbf{g} = \mathbf{F} = 6\pi\mu a\mathbf{U}$ which implies that $\mathbf{U} = \frac{2}{9}\frac{\Delta p a^2}{\mu}\mathbf{g}$ is the terminal velocity for a sphere dropped in a fluid otherwise at rest.

Example. Considering a 1 µm sphere and $\Delta p = 10^3$ kg m^{-3} which gives that $U \approx 2$ µms^{-1} so the Reynolds number is approximately 2×10^{-6} so we can safely ignore inertial forces.

4.3.7 Far-field

For $r \gg a$, the $O\left(\frac{1}{r}\right)$ terms in $\mathbf{u}$ dominate the $O\left(\frac{1}{r^3}\right)$ terms, and the sphere act just like a point exerting a force $\mathbf{F}$ at $\mathbf{x} = \mathbf{0}$. The disturbance flow (to uniform flow or to rest) can be written as

$$\mathbf{u}' = \frac{1}{8\pi\mu}\left(\mathbf{F}\frac{1}{r} + (\mathbf{F}\cdot\mathbf{x})\mathbf{x}\frac{1}{r^3}\right).$$

This is the solution for a point force $\mathbf{f}(\mathbf{x}) = \mathbf{F}\delta(\mathbf{x})$ which we call a Stokeslet.

4.4 Corner Flows

There are many examples of 2D flows in a corner like scraping a fluid. We can scrap the fluid with an inclinded planes moving past each other, or we can hinge two plates together and have them move closer together. We might also have flow going exiting a pipe or flow coming out of a source at the corner of hinged plates. We can also look at the example of flow past a corner which will produce eddies like in a weir.

Sufficiently close to the corner we expect that $L \sim r$ and $\mathrm{Re} \ll 1$ (not true for the source flow since it has $UL \sim Q$ so we need $Q \ll \nu$ to have Stokes flow. We will use a 2D streamfunction ψ with $\mathbf{u} = \nabla \times (0, 0, \psi)$ and plane polar coordinates, so

$$\mathbf{u} = \left(\frac{1}{r} \frac{\partial \psi}{\partial \theta}, -\frac{\partial \psi}{\partial r}, 0 \right)$$

and the vorticity is

$$\boldsymbol{\omega} = \nabla \times \mathbf{u} = (0, 0, -\nabla^2 \psi)$$

and since we know that $\nabla^2 \boldsymbol{\omega} = \mathbf{0}$ in a Stokes flow, so $\nabla^4 \psi = \nabla^2 \nabla^2 \psi = 0$ i.e ψ is biharmonic.

We can get pressure from $\nabla p = \mu \nabla^2 \mathbf{u} = \mu \nabla \times (0, 0, \nabla^2 \psi)$.

Example. (Soucrce flow between plates angled together) Take rigid plates angled at $\theta = \alpha$ and $\theta = -\alpha$ respectively with a source Q at the point where they meet. By mass conservation we will have $u_r \propto \frac{Q}{r}$. So we should try $\psi = Qf(\theta)$ which already gives us that $u_\theta = 0$ so we have purely radial flow, which we expect by the symmetry of the problem. So we have that

$$\nabla^4 \psi = Q \left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \right) \left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \right) f(\theta) = 0$$

which reduces to

$$f'''' + 4f'' = 0$$

and hence we get that $f = A \cos(2\theta) + B \sin(2\theta) + C + D\theta$.

Symmetry implies that u_r has to be even, so ψ is odd so we can remove A and C. So $f(\theta) = B \sin(2\theta) + D\theta$. The no-slip condition gives that $u_r(\pm\alpha) = 0$ which implies that $f'(\pm\alpha) = 0$ so $D = -2B \cos 2\alpha$. For the last condition we need to say something about the flux Q. So we have that

$$Q = \int_0^\infty u_r r \, \mathrm{d}\theta = Q(f(\alpha) - f(-\alpha))$$

and hence $f(\pm\alpha) = \pm\frac{1}{2}$, so we get the solution that

$$f(\theta) = \frac{1}{2} \frac{\sin 2\theta - 2\theta \cos 2\alpha}{\sin 2\alpha - 2\alpha \cos 2\alpha}$$

and finally

$$\mathbf{u} = \frac{Q}{r} \frac{\cos 2\theta - \cos 2\alpha}{\sin 2\alpha - 2\alpha \cos 2\alpha} \mathbf{e}_r.$$

Remark. As $\alpha \to 0$ we see that u_r gains a parabolic profile as expected. The flow is positive only if $2\alpha < \pi$ otherwise if $\alpha > \frac{\pi}{2}$ then we get inflow by the walls.

Remark. if the denominator is zero which happens for $2\alpha = 257^\circ$ we have a problem.

Example. (Moffatt Eddies) Consider the same situation with $Q = 0$ but instead forced by some disturbance at large r outside the local solution. We try a separable solution $\psi = r^\lambda f(\theta)$, so $u \propto r^{\lambda-1}$. Hence we get that

$$\nabla^4 \psi = \nabla^2(r^{\lambda-2}(f'' + \lambda^2 f)) = r^{\lambda-4}(D^2 + (\lambda-2)^2)(D^2+\lambda^2)f = 0$$

where $D = \frac{d}{d\theta}$. We have that $\mathbf{u} = 0$ at $\theta = \pm\alpha$ so $f(\pm\alpha) = f'(\pm\alpha) = 0$, so this an eigenvalue problem for $\lambda(\alpha)$. The general solution (avoiding the special cases where $\lambda = 0, 1, 2$) is

$$f(\theta) = A\cos(\lambda\theta) + B\sin(\lambda\theta) + C\cos(\lambda-2)\theta + D\sin(\lambda-2)\theta.$$

If $f(\theta)$ is even then $B = D = 0$ and u_r is odd and u_θ is even. We have that $f(\pm\alpha) = 0$ so

$$f = A'(\cos(\lambda-2)\alpha\cos(\lambda\theta) - \cos\lambda\alpha\cos(\lambda-2)\theta)$$

. We also have that $f'(\pm\alpha) = 0$ so $\lambda\cos(\lambda-2)\alpha\sin(\lambda\alpha) = (\lambda-2)\cos(\lambda\alpha)\sin(\lambda-2)\alpha$ is an eigenvalue equation for $\lambda(\alpha)$. There is an infinite series of roots, mostly complex, with $\operatorname{Re}\lambda > 1$ so $u \to 0$ in the corner. If $2\alpha \le 146^\circ$ all of the roots are complex. If $\lambda = a \pm ib$ we can write $\psi = r^{a+ib}f(\theta) +$ (complex conjugate solution) to get a real answer. So

$$\psi = r^a(\cos(b\log r)\operatorname{Re} f - \sin(b\log r)\operatorname{Im} f)$$

so we get an infinite number of oscillations as $r \to 0$ so we get a geometric sequence o counter-rotating eddies of decreasing size and strength. The sizes decrease by a factor of $e^{-\pi/b}$ so the velocities decrease by a factor of $e^{-\pi(a-1)/b}$.

If $2\alpha \ge 146^\circ$ then the smallest root (dominant eigenmode) is real so the motion is not oscillatory. We get similar results for f odd which gives $A = C = 0$, which gives oscillator behaviour on the range $2\alpha \le 159^\circ$.

5 Lubrication Theory

5.1 Scaling and Equations

Consider a fluid on a rigid boundary with thickness h which varies along a length scale $L \gg h$ with velocity $\mathbf{u} = (u, v)$. We assume that $\frac{\partial h}{\partial x} \ll 1$ so the fluid surface has small variation.

Incompressibility gives that $\nabla \cdot \mathbf{u} = 0$ so $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$ so

$$\frac{u}{L} \sim \frac{v}{h}$$

and hence $v \ll u$ since $\frac{h}{L} \ll 1$. So the flow is quasi-parallel to the boundary.

For the viscous term, since ∇^2 can be approximated by $\frac{\partial^2}{\partial y^2}$ the viscous term is approximately $\mu\frac{\partial^2}{\partial y^2}$ since $h \ll L$.

In $\rho\frac{Du}{Dt}$ note that $u\frac{\partial}{\partial x} \sim v\frac{\partial}{\partial y} \sim \frac{u}{L}$ and if the geometry is changing, we have that $\frac{\partial}{\partial t} \sim \frac{v}{h} \sim \frac{u}{L}$ from the kinematic boundary condition. So we have that

$$\frac{\rho\frac{Du}{Dt}}{\mu\nabla^2 u} \sim \frac{\rho u^2/L}{\mu u/h^2} \sim \frac{\rho u h^2}{\mu L}.$$

The second approximation is that we assume that

$$\frac{uh}{\nu} \cdot \frac{h}{L} \ll 1$$

and neglect inertia term. This number above is called the *reduced Reynolds number*. The leading-order equation therefore

$$0 = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} + f_x$$

and we expect that $p \sim \frac{\mu u L}{h^2}$ like Poiseuille flow.

We have small flow in the y-direction so

$$\mu \frac{\partial^2 u}{\partial y} \sim \frac{\mu(uh/L)}{h^2} \sim \frac{\mu u}{h^2}\frac{h}{L} \sim \frac{p}{h}\frac{h^2}{L^2} \ll \frac{p}{h}.$$

The leading order equation in the y-direction just says that

$$0 = \frac{\partial p}{\partial y} + f_y.$$

This is the lubrication approximation. This is valid for $h \ll L$, $\frac{\partial h}{\partial x} \ll 1$, $\frac{uh}{\nu}\frac{h}{L} \ll 1$.

We can apply conservation of mass. Write $q(x) = \int u \, \mathrm{d}y$, so we get that

$$(q(x) - q(x + \delta x))\delta t = \delta h \delta x$$

hence taking the δ quantities to zero we get that

$$\frac{\partial h}{\partial t} + \frac{\partial q}{\partial x} = 0.$$

5.2 The Thrust Bearing

Consider the problem of the tooth of a gear a distance d_1 from a solid boundary on the left side and d_2 on the right side in a viscous fluid. We'll let $x = 0$ be the length of tooth and U motion of the gear (but we'll change the frame of reference to have the solid boundary moving in the negative x-direction with velocity U. The geometry gives that $h(x) = d_1 + \alpha x$ where $\alpha = \frac{d_2 - d_1}{L}$. The lubrication approximation is valid is $d_2 \ll L$ (and hence $\alpha \ll 1$ and $\frac{U d_2}{\nu}\frac{d_2}{L} \ll 1$. Then if we have that, since there are no body forces we just have the equations

$$0 = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2}$$

and $0 = -\frac{\partial p}{\partial y}$ with boundary conditions $u = -U$ at $y = 0$ and $u = 0$ at $y = h$. Then we get that

$$u = -\frac{1}{2\mu}\frac{dp}{dx} y(h - y) - U\left(1 - \frac{y}{h}\right)$$

and

$$q = -\frac{h^3}{12\mu}\frac{dp}{dx} - \frac{1}{2}Uh.$$

Mass conversation gives that $\frac{\partial h}{\partial t} = 0$ so $\frac{\partial q}{\partial x} = 0$ so $q = q_0$ independent of x. We assume that the pressure is the same on both sides (i.e the gear is in a large volume of fluid) so we can rearranging the expression for q_0 and integrate. So

$$\begin{aligned} \frac{dp}{dx} &= -12\mu \left(\frac{q_0}{h^3} + \frac{U}{2h^2} \right) \\ \implies p(x) &= p_0 + \frac{6\mu}{\alpha} \left(q_0 \left(\frac{1}{h^2} - \frac{1}{d_1^2} \right) + U \left(\frac{1}{h} - \frac{1}{d_1} \right) \right) \end{aligned}$$

but we have that $p(L) = 0$, so we must have that

$$q_0 = -U \frac{d_1 d_2}{d_1 + d_2}.$$

This allows us to calculate $p(x)$, $u(x, y)$ and the pressure gradient.

Remark. There must be a point between $x = 0$ and $x = L$ where the pressure is at its maximum and $\frac{dp}{dx} = 0$. This occurs at

$$h = \frac{2}{\frac{1}{d_1} + \frac{1}{d_2}}$$

the harmonic mean and there the flow profile is just Couette.

The force on the bottom plate, with $\mathbf{n} = (0, 1)$ can be calculated using the normal and tangential components.

$$\begin{aligned} F_{\mathrm{N}} &= -\int_0^L (p - p_0) \, \mathrm{d}x = -\frac{6\mu U}{\alpha^2} \left(\log \frac{d_2}{d_1} - 2 \frac{d_2 - d_1}{d_2 + d_1} \right) \\ F_{\mathrm{T}} &= \int_0^L \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \mathrm{d}x = \frac{4\mu U}{\alpha} \left(\log \frac{d_2}{d_1} - \frac{3}{2} \frac{d_2 - d_1}{d_2 + d_1} \right). \end{aligned}$$

So if $d_2 - d_1 \sim d_1$ then the bracketed terms are $O(1)$ and we have that $\alpha \sim \frac{d_1}{L}$ so

$$F_{\mathrm{N}} \sim \frac{\mu U L}{d_1^2} \cdot L \qquad F_{\mathrm{T}} \sim \frac{\mu U}{d_1} \cdot L.$$

Hence we can support a large load/thrust F_{N} with friction F_{T} moderate to low. Note that $d_1 \sim \left(\frac{F_{\mathrm{N}}}{\mu u} \right)^{\frac{12}{L}}$ so choose oil sufficiently viscous that d_1 is greater than the roughness on the plate. We have normal and tangential vectors $\mathbf{n} \propto \left(\frac{dh}{dx}, -1 \right)$ and $\mathbf{t} \propto \left(1, \frac{dh}{dx} \right)$. From the general scalings we have that

$$\sigma = -pI + \begin{pmatrix} 0 & \mu \frac{\partial u}{\partial y} \\ \mu \frac{\partial u}{\partial y} & 0 \end{pmatrix}$$

where the first term is $O\left(\frac{\mu u L}{h^2} \right)$ and the second term is $O\left(\frac{\mu u}{h} \right)$ with additional correction terms being $O\left(\frac{\mu u}{L} \right)$.

At leading order, $\mathbf{n} \cdot \sigma \cdot \mathbf{n} = -p$ hence $F_{\mathrm{N}} \approx F_y$. We also have that $\mathbf{t} \cdot \sigma \cdot \mathbf{n} = -\mu \left. \frac{\partial u}{\partial y} \right|_{y=h}$ which is the tangential stress.

Remark. Intuitively this is simple however $F_{\mathrm{T}} \neq F_x$ even at leading order, beause $\mathbf{e}_x \cdot \sigma \cdot \mathbf{n} = -\mu \left. \frac{\partial u}{\partial y} \right|_{y=h} - p \frac{dh}{dx}$, where the second term is $O\left(\frac{\mu u}{h} \right)$ (the same order as the first term).

The journal bearing is the same problem as this, but we wrap it around a circle with a changing gap thickness $h(\theta)$ between the two cylinders. Take local coordinates $y = r - a$ and $x = a\theta$ so we get the same leading order equations locally and impose the condition that

$$\int_0^{2\pi} \frac{dp}{d\theta}\, \mathrm{d}\theta = 0$$

to find the flux q_0. The pressure difference from opposite sides supports the load and prevents the axel from coming into contact with the opposite cylinder.

5.3 Cylinder Approaching a Wall

Suppose we have a cylinder approaching a wall with the wall at $y = 0$ and the x axis parallel to the wall's rigid surface. Let $h(x)$ be the gap between the cylinder and the wall and the cylinder is in a large tank of fluid. The cylinder has radius a and velocity $-V\mathbf{e}_y$. Let d be the minimum gap thinkness and assume that $d \ll a$ and $\frac{uh}{\nu}\frac{h}{L} \sim \frac{Vh}{\nu} \ll 1$.

We have that $h - d = a(1 - \cos\theta)$ where θ is the angle subtended from the center of the cylinder at x, so $x = a\sin\theta$. So

$$h \approx d + a\frac{\theta^2}{2}, \qquad x \approx a\theta$$

which together gives that

$$h \approx d\left(1 + \frac{x^2}{2ad}\right).$$

Therefore the relevant lengthscale is $L \sim \sqrt{ad}$, with $d \ll \sqrt{ad} \ll a$. So we do have a long thin flow, hence we can apply lubrication theory. Since we have that $\sqrt{ad} \ll a$ our parabolic approximation is all good.

Now we look at the local quasiparallel with

$$\mu\frac{\partial^2 u}{\partial y^2} = \frac{\partial p}{\partial x}, \quad u(0) = u(h) = 0$$

which gives that

$$u = -\frac{1}{2\mu}\frac{\partial p}{\partial x}y(h - y) \implies q = -\frac{h^3}{12\mu}\frac{\partial p}{\partial x}.$$

Now we can apply mass conservation. Physically it's obvious that $q = Vx$ and mathemaically

$$\frac{\partial q}{\partial x} = -\frac{\partial h}{\partial t} = V \implies q = Vx$$

gives the same result. We have that

$$\frac{\partial p}{\partial x} = -\frac{12\mu Vx}{d^3\left(1 + \frac{x^2}{2ad}\right)^3} \implies p = p_0 + \frac{6\mu Va}{d^2\left(1 + \frac{x^2}{2ad}\right)^2}$$

with the constant of integration set to p_0 since the right term decays rapidly on lengthscale L, so we can impose $p \to p_0$ as $x \to \infty$ where x is much larger than d but not much larger than a i.e on the L lengthscale.

Now let's look at the vertical force on the cylinder per unit length into the page.

$$F = \int_{-\infty}^{\infty} (p - p_0)\ \mathrm{d}x = \frac{6\mu V a}{d^2}\sqrt{2ad}\int_{-\infty}^{\infty} \frac{\mathrm{d}t}{(1+t^2)^2}$$
$$= \frac{3\pi\mu V a\sqrt{2ad}}{d^2}.$$

Note that $V = -\dot{d}$ so if the cylinder is settling under its own weight, so $F = \Delta\rho g\pi a^2$, the weight is constant so $\frac{\dot{d}}{d^{3/2}}$ is constant, hence $d \sim t^{-2}$ as $t \to \infty$ so there is infinite time to contact. We can rewrite

$$F = 6\pi\mu a V\left(\frac{a}{2d^3}\right)^{3/2}$$

so we can see the force here is much bigger than if the cylinder was just settling on its own in a fluid.

5.4 Hele-Shaw Flow

Take two long plates with a constant width h apart with an induced flow of velocity U on one side of the plate. We introduce an obstruction between the plates with lengthscale L. Take the plates in the xy plane and the velocity coming in so it is positive in the x direction. Assume that $h \ll L$ so the flow is mainly in the xy plane and $\mathbf{u} = (u, v, w)$ with $w = O\left(\frac{h}{L}u, \frac{h}{L}v\right)$. Assume that interial forces are negliable, so

$$\frac{(u,v)h}{\nu}\frac{h}{L} \ll 1$$

hence we can apply lubrication theory. So we get that

$$\frac{\partial p}{\partial z} = 0, \quad \mu\frac{\partial^2 \mathbf{u}_\mathrm{H}}{\partial z^2} = \nabla_\mathrm{H} p$$

where $\mathbf{u}_\mathrm{H} = (u, v)$ and $\nabla_\mathrm{H} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right)$.

We have that

$$\mathbf{u}_\mathrm{H} = \frac{1}{2\mu}\nabla_\mathrm{H} p\ z(h-z) \implies \mathbf{q}_\mathrm{H} = \int_0^h \mathbf{u}_\mathrm{H}\ \mathrm{d}z = -\frac{h^3}{12\mu}\nabla_\mathrm{H} p.$$

From mass conservation, $\nabla \cdot \mathbf{u} = 0$ so $\nabla_\mathrm{H} \cdot \mathbf{u}_\mathrm{H} + \frac{\partial w}{\partial z} = 0$. So integrating with resepct to z (since w evaulated at 0 and h is zero, we get that $\nabla_\mathrm{H} \cdot \mathbf{q}_\mathrm{H} = 0$. So

$$\mathbf{q}_\mathrm{H} = -\frac{h^3}{12\mu}\nabla_\mathrm{H} p \implies \nabla_\mathrm{H}^2\ p(x,y) = 0.$$

This looks like IB potential flow where $\mathbf{u} = \nabla\phi$ and $\nabla^2\phi = 0$. Therefore the viscous flow in a Hele-Shaw cell gives a visualation of 2D potential flow.

Example. Consider uniform flow past a cylinder. As $r \to \infty$ we have that $\mathbf{q} \to Q\mathbf{e}_x$ which implies that

$$p = -\frac{12\mu Q}{h^3} r\cos\theta.$$

On $r = a$ we have that $\mathbf{q} \cdot \mathbf{n} = 0$ which gives that $\frac{\partial p}{\partial r} = 0$. Further we have that $\nabla_\mathrm{H}^2\ p = 0$.

5.5 Gravitational Spreading of a Viscous Fluid

Assume we have axisymmetric fluid with height $z = h(r,t)$ and radius $r = r_N(t)$. We'll also assume inertia is negligible (so the reduced Reynolds number is $\ll 1$, along with neglecting any surface tension and issues at the moving contact line, and assume the surface is long and thin so $h \ll r_N(t)$. Now we can use lubrication theory. How fast does thie viscous gravity current spread? Vertical balance gives that

$$\begin{cases} 0 = -\frac{\partial p}{\partial z} - \rho g \\ p = p_0 \text{ at } z = h \end{cases} \implies p = p_0 + \rho g(h - z).$$

Flow in the radial direction gives that

$$0 = -\frac{\partial p}{\partial r} + \mu \frac{\partial^2 u}{\partial z^2}, \quad u = 0 \text{ at } z = 0 \text{ and } \mu \frac{\partial u}{\partial z} = 0 \text{ at } z = h.$$

Now $-\frac{\partial p}{\partial r} = -\rho g \frac{\partial h}{\partial r}$ independent of z implies that

$$u_r = -\frac{1}{2\mu} \frac{\partial p}{\partial r} z(2h - z)$$

so we get that

$$q = \int_0^h u \, \mathrm{d}z = -\frac{g}{3\nu} h^3 \frac{\partial h}{\partial r}.$$

Now we can apply mass conservation. By considering a small change in the radius δr we have that

$$2\pi r q(r) - 2\pi(r + \delta r) q(r + \delta r) = 2\pi r \delta r \frac{\partial h}{\partial t}$$

to leading order. Hence we get that

$$\frac{\partial h}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r}(rq) = 0 \tag{*}$$

so applying to our q we get that

$$\frac{\partial h}{\partial t} = \frac{g}{3\nu} \frac{1}{r} \frac{\partial}{\partial r} \left(r h^3 \frac{\partial h}{\partial r} \right).$$

No source at $r = 0$ implies that $2\pi r q \to 0$ as $r \to 0$ and $h = 0$ at $r = r_N$. Now $\int (*) r \mathrm{d}r$ and the other two conditions gives that $\int hr \, \mathrm{d}r = V$, constant volume. Now since $(*)$ is a nonlinear diffusion equation, we can't solve it analytically. Instead we have to solve it numerically. We find that its solution quickly forgets about the initial conditions, other than the initial volume V, and tends towards a similarity solution of the form

$$h(r,t) = H(t) f(\eta)$$

where $\eta = \frac{r}{R(t)}$. Let's solve for H and f. We solve by scaling then substitute to find the shape $f(\eta)$. We have a constant volume constraint, so $HR^2 \sim V$ and a governing time evolution equation

$$\frac{H}{t} \sim \frac{g}{\nu} \frac{H^4}{R^2} \sim \frac{g}{\nu} \frac{H^5}{V}$$

and set $H(t) = \left(\frac{\nu V}{gt}\right)^{\frac{1}{4}}$ and $R(t) = \left(\frac{gtV^3}{\nu}\right)^{\frac{1}{8}}$ and $\eta = \frac{r}{R(t)}$ and substitute into the governing PDE, noting that $H(t) \propto t^{-1/4}$ so $\frac{dH}{dt} = -\frac{1}{4}\frac{H}{t}$. Similarly $\eta \propto t^{-1/8}$ so $\frac{\partial \eta}{\partial t} = -\frac{1}{8}\frac{\eta}{t}$ and $\frac{\partial}{\partial r} = \frac{1}{R}\frac{\partial}{\partial \eta}$. We have that

$$-\frac{1}{4}\frac{H}{t}f + H\frac{df}{d\eta}\left(-\frac{1}{8}\frac{\eta}{t}\right) = \frac{g}{3\nu}\frac{H^4}{R^2}\frac{1}{\eta}\cdot \underbrace{\frac{d}{d\eta}\left(\eta f^3 \frac{df}{d\eta}\right)}_{\text{do not expand this term}}.$$

Now we have that $\frac{H}{t} = \frac{g}{\nu}\frac{H^4}{R^2}$ by design which gives that

$$-\frac{1}{4}f - \frac{1}{8}\eta f' = \frac{1}{3}\frac{1}{\eta}(\eta f^3 f')'$$

which is an ODE for $f(\eta)$. Similarly the volume constraint becomes

$$2\pi \int_0^{\eta_N} f\eta \, \mathrm{d}\eta = 1$$

where $\eta_N = \frac{r_N}{R}$. To solve the ODE, multiply by η and we can then integrate so we get that

$$-\frac{1}{8}\eta^2 f = \frac{1}{3}\eta f^3 f + \text{const.}$$

Now we can use the fact that f' finite at $\eta = 0$ since we don't have any source flux. So this gives that $\frac{1}{3}f^2 f' = -\frac{1}{8}\eta$. Hence $\frac{1}{9}f^3 = \frac{1}{16}(\eta_N^2 - \eta^2)$. Choose constant so that $f(\eta_N) = 0$ hence

$$f(\eta) = \left[\frac{9}{16}(\eta_N^2 - \eta^2)\right]^{\frac{1}{3}}.$$

We can find η_N frmo the volume constraint integral.

$$12\pi\left(\frac{9}{16}\right)^{\frac{1}{3}} \int_0^{\eta_N} (\eta_N^2 - \eta^2)^{\frac{1}{3}}\eta \, \mathrm{d}\eta = \frac{3}{8}\eta_N^{8/3}.$$

So we get that

$$\eta_N = \left(\frac{2^{10}}{3^5\pi^3}\right)^{\frac{1}{8}}.$$

Gravity current spreads with

$$r_N = \eta_N\left(\frac{V^3 gt}{\eta}\right)^{\frac{1}{8}}$$

which is very slow since $r_N \propto t^{\frac{1}{8}}$. The 2D case is on Example Sheet 3. The boundary layer equations also have a similarity solution, but usually the nonlinear ODE for $f(\eta)$ must be solved numerically.

6 Vorticity Generation and Confinement

6.1 The Vorticity Equation

Starting now with the full incompressible Navier-Stokes equation and assume that body forces are conservative, so $\mathbf{f} = -\nabla\chi$. Hence

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u} - \nabla\chi.$$

Recall from IB Fluid Dynamics that

$$\mathbf{u} \times (\nabla \times \mathbf{u}) = \nabla \left(\frac{1}{2} u^2 \right) - (\mathbf{u} \cdot \nabla)\mathbf{u}$$

so

$$\begin{aligned} \nabla \times ((\mathbf{u} \cdot \nabla)\nabla) &= -\nabla \times (\mathbf{u} \times \boldsymbol{\omega}) \\ &= -(\mathbf{u}\nabla \cdot \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \nabla \mathbf{u} - \boldsymbol{\omega}\nabla \cdot \mathbf{u} - \mathbf{u} \cdot \nabla \boldsymbol{\omega}) \\ &= \mathbf{u} \cdot \nabla \boldsymbol{\omega} - \boldsymbol{\omega} \cdot \nabla \mathbf{u}. \end{aligned}$$

Hence for ρ, μ uniform, taking the curl of Navier-Stokes we have that

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + \underbrace{(\mathbf{u} \cdot \nabla)\boldsymbol{\omega}}_{\text{advection}} = \underbrace{(\boldsymbol{\omega} \cdot \nabla)\mathbf{u}}_{\text{vortex stretching}} + \underbrace{\nu \nabla^2 \boldsymbol{\omega}}_{\text{diffusion}}.$$

Remark. We have no source terms, so if $\boldsymbol{\omega} = 0$ in the volume then it is zero for all time, unless it comes from the boundary.

Remark. In 2D flow, $\mathbf{u} = (\psi_y, -\psi_x, 0)$ so $\boldsymbol{\omega} = (0, 0, \omega)$ so $(\boldsymbol{\omega} \cdot \nabla)\mathbf{u} = \mathbf{0}$, so no vortex stretching in a 2D flow.

6.2 Vortex Stretching

Recall from IB Fluid Dynamics that a material line element $\boldsymbol{\delta \ell}$ is advected according to

$$\frac{d}{dt} \boldsymbol{\delta \ell} = (\boldsymbol{\delta \ell} \cdot \nabla)\mathbf{u}.$$

If its diffusion is negligible then vorticity is advected, rotated and stretched in just the same way as a material line ement with $\boldsymbol{\delta \ell}$ parallel to $\boldsymbol{\omega}$. Note that

$$(\boldsymbol{\omega} \cdot \nabla)\mathbf{u} = \boldsymbol{\omega} \cdot e \implies \boldsymbol{\omega} \times \boldsymbol{\omega} = 0$$

i.e the vorticity doesn't rotate itself: it's is only rotated by straining type motion. Steady strain amplifies $\boldsymbol{\omega}$ in the direction(s) of stretching.